



Smart Vehicle Safety & Alert System

Abstract

Introduction

Personal safety during travel has become an important concern for individuals travelling alone in their personal vehicles. Situations such as being followed, road rage accidents, or travelling on isolated roads can create emergencies where immediate help is required. In such cases, relying only on mobile phone may not be effective, especially during long journeys or stressful situations.

Problem Statement

Most existing safety solutions depend on smartphones and internet connectivity to send emergency alerts. However, mobile phones may run out of battery, lose network coverage, or become inaccessible during critical situations. There is a need for a dedicated, vehicle-mounted safety system that allows anyone travelling alone to quickly alert emergency contacts without depending on mobile phones or internet services.

Conclusion

To address this problem, the **Smart Vehicle Safety & Alert System** is developed as a B2C-based personal safety solution. The system includes a discreet SOS button integrated inside the vehicle, which allows the user to manually trigger an emergency alert when required. Once activated, the system retrieves the real-time location of the vehicle using a GPS module and sends alert message through a GSM module to pre-stored emergency contacts. Controlled by an Arduino micro controller, the system operates independently of smartphones, and internet connectivity. This project provides a low-cost, reliable, and effective personal safety solution that enhances security and ensures timely assistance for individuals during emergencies.

Guide Name: Mr.Srinu Naik

Batch no: 16

23911A6685 Ch. Surya

Prashanthi

23911A66B3 P. Kushal Babu

23911A66B8 S.Siri

24915A6612 M.Akshath Sai

Signature